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Time-to-Event Prediction with Neural Networks and Cox Regression

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Abstract

New methods for time-to-event prediction are proposed by extending the Cox proportional hazards model with neural networks. Building on methodology from nested case-control studies, we propose a loss function that scales well to large data sets and enables fitting of both proportional and non-proportional extensions of the Cox model. Through simulation studies, the proposed loss function is verified to be a good approximation for the Cox partial log-likelihood. The proposed methodology is compared to existing methodologies on real-world data sets and is found to be highly competitive, typically yielding the best performance in terms of Brier score and binomial log-likelihood. A python package for the proposed methods is available at `https://github.com/havakv/pycox`.

Keywords: Cox regression, customer churn, neural networks, non-proportional hazards, survival prediction

1. Introduction

In this paper, we consider methodology for time-to-event prediction, a part of survival analysis that reasons about when a future event will occur. Applications of time-to-event predictions can be found in a variety of settings such as survival prediction of cancer patients (e.g., Vigan et al., 2000), customer churn (e.g., Van den Poel and Lariviere, 2004), credit scoring (e.g., Dirick et al., 2017), and failure times of mechanical systems (e.g., Susto et al., 2015). Arguably, the field of survival analysis has predominantly focused on interpretability, potentially at some cost of predictive accuracy. This is perhaps the reason why binary classifiers from machine learning are commonly used in industrial applications where survival methodology is applicable. However, while the binary classifiers can provide predictions for *one* predetermined duration, one loses the interpretability and flexibility provided by modeling the event probabilities as a function of time. Furthermore, in time-to-event data, it is common that some individuals are not followed all the way to their event time, resulting in *censored* times rather than event times. While binary classifiers typically ignore these observations, one of the main objectives in survival analysis is to account for them. Hence,

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in applications with a substantial amount of censoring, the use of survival models tends to be advantageous.

In our work, we propose an approach for combining machine learning methodology with survival models. We do this by extending the Cox proportional hazards model with neural networks, and further remove the proportionality constraint of the Cox model. Building on methodology from nested case-control studies (e.g., Langholz and Goldstein, 1996) we are able to do this in a scalable manner. The resulting methods have the flexibility of neural networks while modeling event times continuously. Building on the PyTorch framework (Paszke et al., 2017), we provide a python package for our methodology, along with all the simulations and data sets presented in this paper.[1]

The paper is organized as follows. Section 2 contains a summary of related work. In Section 3, we review some basic concepts from survival analysis and introduce the Cox proportional hazards model with our extensions. In Section 4 we discuss some evaluation criteria of methods for time-to-event prediction. In Section 5, we conduct a simulation study, verifying that the methods we propose behave as expected. In Section 6 we evaluate our methods on five real-world data sets and compare their performances with existing methodology. We conclude in Section 7.

2. Related Work

The extension of Cox regression with neural networks was first proposed by Faraggi and Simon (1995), who replaced the linear predictor of the Cox regression model, cf. formula (3) below, by a one hidden layer multilayer perceptron (MLP). It was, however, found that the model generally failed to outperform regular Cox models (Xiang et al., 2000; Sargent, 2001). Katzman et al. (2018) revisited these models in the framework of deep learning and showed that novel networks were able to outperform classical Cox models in terms of the C-index (Harrell Jr et al., 1982). Our work distinguishes itself from this in the following way: The method by Katzman et al. (2018), denoted DeepSurv, is constrained by the proportionality assumption of the Cox model while we propose an extension of the Cox model where proportionality is no longer a restriction. In this regard, we propose an alternative loss function that scales well for both the proportional and the non-proportional cases.

Similar works based on Cox regression include SurvivalNet (Yousefi et al., 2017), a framework for fitting proportional Cox models with neural networks and Bayesian optimization of the hyperparameters, and Zhu et al. (2016) and Zhu et al. (2017) which extended the Cox methodology to images. Both Zhu et al. (2016) and Zhu et al. (2017) replace the MLP of DeepSurv with a convolutional neural network and applied these methods to pathological images of lung cancer and to whole slide histopathological images.

An alternative approach to time-to-event prediction is to discretize the duration and compute the hazard or survival function on this predetermined time grid. Luck et al. (2017) proposed methods similar to DeepSurv, but with an additional set of discrete outputs for survival predictions and computed an isotonic regression loss over this time grid. Fotso (2018) parameterized a multi-task logistic regression with a neural net that directly computes the survival probabilities on the time grid. Lee et al. (2018) proposed a method,

1. Implementations of methods and the data sets are available at `https://github.com/havakv/pycox`.

denoted DeepHit, that estimates the probability mass function with a neural net and combine the log-likelihood with a ranking loss; see Appendix D for details. Furthermore, the method has the added benefit of being applicable for competing risks.

The majority of the papers mentioned benchmark their methods against the random survival forests (RSF) by Ishwaran et al. (2008). RSF computes a random forest using the log-rank test as the splitting criterion. It computes the cumulative hazards of the leaf nodes and averages them over the ensemble. Hence, RSF is a very flexible continuous-time method that is not constrained by the proportionality assumption.

3. Methodology

In the following, we give a brief review of some concepts in survival analysis. For a more in-depth introduction to the field, see, for example, Klein and Moeschberger (2003).

Our objective is to model the event distribution as a continuous function of time. So with $f(t)$ and $F(t)$ denoting the probability density function and the cumulative distribution function of an event time T^*, we want to model

$$\mathrm{P}(T^* \leq t) = \int_0^t f(s)\, ds = F(t).$$

As alternatives to $F(t)$, it is common to study the *survival function* $S(t)$ and the *hazard rate* $h(t)$. The survival function is defined as

$$S(t) = \mathrm{P}(T^* > t) = 1 - F(t),$$

and is commonly used for visualizing event probabilities over time. For specifying models, however, it is rather common to use the hazard rate

$$h(t) = \frac{f(t)}{S(t)} = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \mathrm{P}(t \leq T^* < t + \Delta t \mid T^* \geq t).$$

If we have the hazard rate, the survival function can be retrieved through the cumulative hazard, $H(t) = \int_0^t h(s)ds$, by

$$S(t) = \exp[-H(t)]. \tag{1}$$

The survival function and the hazard rate therefore provide contrasting views of the same quantities, and it may be useful to study both.

Working with real data, the true event times are typically not known for all individuals. This can occur when the follow-up time for an individual is not long enough for the event to happen, or an individual may leave the study before its termination. Instead of observing the true event time T^*, we then observe a possibly right-censored event time $T = \min\{T^*, C^*\}$, where C^* is the censoring time. In addition, we observe the indicator $D = \mathbb{1}\{T = T^*\}$ labeling the observed event time T as an event or a censored observation. Now, denoting individuals by i, with covariates $\mathbf{x}_i$ and observed duration T_i, the likelihood for censored survival times is given by

$$L = \prod_i f(T_i \mid \mathbf{x}_i)^{D_i} S(T_i \mid \mathbf{x}_i)^{1-D_i} = \prod_i h(T_i \mid \mathbf{x}_i)^{D_i} \exp[-H(T_i \mid \mathbf{x}_i)]. \tag{2}$$

We will later refer to this as the *full likelihood.*

3.1 Cox Regression

The Cox proportional hazards model (Cox, 1972) is one of the most used models in survival analysis. It provides a semi-parametric specification of the hazard rate

$$h(t \mid \mathbf{x}) = h_0(t) \exp[g(\mathbf{x})], \qquad g(\mathbf{x}) = \boldsymbol{\beta}^T \mathbf{x}, \tag{3}$$

where $h_0(t)$ is a non-parametric *baseline hazard*, and $\exp[g(\mathbf{x})]$ is the *relative risk function*. Here, $\mathbf{x}$ is a covariate vector and $\boldsymbol{\beta}$ is a parameter vector. Note that the linear predictor $g(\mathbf{x}) = \boldsymbol{\beta}^T \mathbf{x}$ does not contain an intercept term (bias weight). This is because the intercept would simply scale the baseline hazard, and therefore not contribute to the relative risk function,

$$h_0(t) \exp[g(\mathbf{x}) + b] = h_0(t) \exp[b] \exp[g(\mathbf{x})] = \tilde{h}_0(t) \exp[g(\mathbf{x})].$$

The Cox model in (3) is fitted in two steps. First, the parametric part is fitted by maximizing the *Cox partial likelihood*, which does not depend on the baseline hazard, then the non-parametric baseline hazard is estimated based on the parametric results. For individual i, let T_i denote the possibly censored event time and $\mathcal{R}_i$ denote the set of all individuals at risk at time T_i (not censored and have not experienced the event before time T_i). Note that $\mathcal{R}_i$ includes individuals with event times at T_i, so i is part of $\mathcal{R}_i$. The Cox partial likelihood, with Breslow's method for handling tied event times, is given by

$$L_{\text{cox}} = \prod_i \left(\frac{\exp[g(\mathbf{x}_i)]}{\sum_{j \in \mathcal{R}_i} \exp[g(\mathbf{x}_j)]} \right)^{D_i}, \tag{4}$$

and the negative partial log-likelihood can then be used as a loss function

$$\text{loss} = \sum_i D_i \log \left(\sum_{j \in \mathcal{R}_i} \exp[g(\mathbf{x}_j) - g(\mathbf{x}_i)] \right). \tag{5}$$

Let $\hat{\boldsymbol{\beta}}$ be the value of $\boldsymbol{\beta}$ that maximizes (4), or equivalently, minimizes (5). Then the cumulative baseline hazard function can be estimated by the *Breslow estimator*

$$\begin{aligned} \hat{H}_0(t) &= \sum_{T_i \leq t} \Delta \hat{H}_0(T_i) \\ \Delta \hat{H}_0(T_i) &= \frac{D_i}{\sum_{j \in \mathcal{R}_i} \exp[\hat{g}(\mathbf{x}_j)]}, \end{aligned} \tag{6}$$

where $\hat{g}(\mathbf{x}) = \hat{\boldsymbol{\beta}}^T \mathbf{x}$. If desired, the baseline hazard $h_0(t)$ can be estimated by smoothing the increments, $\Delta \hat{H}_0(T_i)$, of the Breslow estimate, but the cumulative baseline hazard typically provides the information we are interested in.

3.2 Cox with SGD

The Cox partial likelihood is usually minimized using Newton-Raphson's method. In our work, we instead want to fit the Cox model with mini-batch stochastic gradient descent